

Subgroups:

Definition: H of G is a subset of G satisfying the group axioms

Two trivial subgroups: $\{e\}$ and G

If $g \in G$, the smallest subgroup containing g is

$$\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}.$$

Ex: D_3

$$\{e\}, \quad D_3, \quad \{e, r, r^2 = r^{-1}\}, \quad \{e, \sigma_1\}.$$

Normal subgroup:

A subgroup H is normal (invariant) if

$$gHg^{-1} = H, \quad \forall g \in G.$$

$$\boxed{H \triangleleft G}$$

Ex: In D_3 : $\{e, r, r^{-1}\}$ is the only proper normal subgroup.

Simple and Semi-simple groups:

Simple: No nontrivial normal subgroups

Semi-Simple: No abelian normal subgroups

Coset decomposition and Lagrange's theorem

H is a subgroup of G . Cosets are a way of partitioning the group into disjoint pieces of equal size, each piece being “a shifted copy” of the subgroup.

For $\forall a \in G$,

$$Ha = \{ha \mid h \in H\}, \quad \text{right coset}$$

$$aH = \{ah \mid h \in H\}, \quad \text{left coset}$$

Property: $Ha_i = Ha_j$ or $Ha_i \cap Ha_j = \emptyset$

Property: If H is normal,

$$aHa^{-1} = H,$$

$$aH = Ha,$$

all left and right coset coincide ...

Lagrange theorem

Let G be a finite group, H a subgroup. Then:

1.

$$G = Ha_1 \cup Ha_2 \cup \cdots \cup Ha_n, \quad a_1 = e,$$

$$G = b_1H \cup b_2H \cup \cdots \cup b_nH, \quad b_1 = e.$$

2.

$$|Ha_i| = |a_iH| = |H|.$$

3. The number of cosets is the index of H in G :

$$|G| = n|H|, \quad n = |G : H|.$$

Ex:

$$\begin{aligned} D_3 &= \{e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3\}, & |D_3| &= 6, \\ H &= \{e, \sigma_1\}, & |H| &= 2 \\ \implies n &= |G : H| = \frac{6}{2} = 3. \end{aligned}$$

Let's compute left cosets

$$\begin{aligned} eH &= \{e, \sigma_1\} = H \\ rH &= \{re, r\sigma_1\} = \{r, \sigma_3\} \\ r^{-1}H &= \{r^{-1}, r^{-1}\sigma_1\} = \{r^{-1}, \sigma_2\} \\ \sigma_2H &= \{\sigma_2, \sigma_2\sigma_1\} = r^{-1}H \\ \sigma_3H &= rH \\ \sigma_1H &= eH \end{aligned}$$

$$\begin{aligned} D_3 &= \{e, \sigma_1\} \cup \{r, \sigma_3\} \cup \{r^{-1}, \sigma_2\} \\ \implies & \text{ for a right coset:} \\ D_3 &= \{e, \sigma_1\} \cup \{r, \sigma_2\} \cup \{r^{-1}, \sigma_3\}. \end{aligned}$$

Exercise

$$\begin{aligned} K &= \{e, r, r^{-1}\}, & gKg^{-1} &= K, \quad \forall g \in D_3 \\ gK &= Kg \\ |K| &= 3 \implies n = \frac{6}{3} = 2. \end{aligned}$$

$$srs = r^{-1}, \quad srs^{-1} = r^{-1}, \quad r^3 = e.$$

$$\begin{aligned} eK &= K \\ rK &= \{re, rr, rr^{-1}\} = \{r, r^2, e\} = K \\ r^2K &= \{r^2e, r^2r, r^2r^{-1}\} = \{e, r, r^2\} = K \\ \sigma_1K &= \{\sigma_1e, \sigma_1r, \sigma_1r^2\} = \{\sigma_1, \sigma_3, \sigma_2\} \\ &\dots \\ \sigma_2K &= \sigma_3K = \sigma_1K = \{\sigma_1, \sigma_2, \sigma_3\} \\ D_3 &= K \cup \underbrace{\sigma_1K}_{\text{rotations}} \cup \underbrace{\sigma_1K}_{\text{reflections}}. \end{aligned}$$

Conjugate class decomposition

Conjugate elements: $a, b \in G$; b is conjugate of a if there exists $g \in G$ such that

$$b = gag^{-1}.$$

Prop:

$$1. \ a = eae^{-1}.$$

$$2. \ \text{If } b = gag^{-1},$$

$$a = g^{-1}bg.$$

$$3. \ \text{If } b = g_1ag_1^{-1}, \ c = g_2bg_2^{-1},$$

$$c = g_2g_1a(g_2g_1)^{-1}.$$

Conjugacy classes:

The conjugacy class of an element $a \in G$ is the set

$$(a) = \{gag^{-1} \mid g \in G\}.$$

Prop:

1. If $(a) = \{a\}$, then a is self-conjugate.
2. In any abelian groups, every element is self-conjugate

$$[gag^{-1} = a].$$

3. Conjugacy classes are invariant under conjugation:

$$g(a)g^{-1} = (a), \quad \forall g \in G.$$

Ex: D_3

$$(e) = \{e\} \iff \text{exercise}$$

$$(r) = (r^{-1}) = \{r, r^{-1}\}$$

$$(\sigma_1) = (\sigma_2) = (\sigma_3) = \{\sigma_1, \sigma_2, \sigma_3\}$$

Theorem: Every group G is the disjoint union of the conjugacy classes of its elements

$$G = \bigcup_{a \in G} (a)$$

$$D_3 = \{e\} \cup \{r, r^{-1}\} \cup \{\sigma_1, \sigma_2, \sigma_3\}.$$

Quotient group: H is a normal subgroup of G ($H \triangleleft G$)

Assume

$$G = a_1H \cup a_2H \cup \dots \cup a_nH, \quad a_1 = e.$$

The set

$$G/H = \{a_1H, a_2H, \dots, a_nH\}$$

is called the quotient group of G by H .

Prop:

1. For $a_iH, a_jH \in G/H$ the composition is defined as

$$(a_iH)(a_jH) = (a_ia_j)H.$$

2. $eH = He = H$.

- 3.

$$(a_iH)^{-1} = a_i^{-1}H.$$

4. $|G| = n_G, |H| = n_H$

$$\implies |G/H| = \frac{n_G}{n_H}.$$

Ex: D_3

$$H = \{e, r, r^{-1}\}$$

$$eH = rH = r^{-1}H$$

$$\sigma_1H = \sigma_2H = \sigma_3H = \{\sigma_1, \sigma_2, \sigma_3\}$$

$$D_3/H = \{H, \sigma_1H\}.$$

D_3/H	H	$\sigma_1 H$
H	$(H)(H) = H$	$(H)(\sigma_1 H) = \sigma_1 H$
$\sigma_1 H$	$(\sigma_1 H)(H) = \sigma_1 H$	$(\sigma_1 H)(\sigma_1 H) = \sigma_1 \sigma_1 H = H$

$$C_2 = \{e, g\}, \quad g^2 = e, \quad D_3/H \simeq C_2.$$

Centraliser:

The centraliser of S in G is the set of

$$C_G(S) = \{g \in G \mid gs = sg \text{ for all } s \in S\}.$$

Normalizer:

$$N_G(S) = \{g \in G \mid gSg^{-1} = S\} = \{g \in G \mid gsg^{-1} \in S \text{ for all } s \in S\}.$$

Center:

Center of G :

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Remarks

1. $C_G(S)$ and $N_G(S)$ are subgroups of G .
2. $Z(G)$ is always abelian subgroup of G .
3. If S is itself a subgroup then $C_G(S)$ measures how close S is to being central – whereas $N_G(S)$: how close S is to being normal.

Direct product

A group G is the direct product of H and K , $G = H \times K$ if:

1. Every $h \in H$ commutes with $k \in K$, $hk = kh$.
2. Every $g \in G$ can be written uniquely as $g = hk$.

Prop.

$$\begin{aligned} |H| &= n_H, & |K| &= n_K \\ \implies |G| &= n_H \times n_K. \end{aligned}$$

Composition

$$(h_1, k_1)(h_2, k_2) = (h_1 h_2, k_1 k_2).$$

Ex:

$$\begin{aligned} G &= D_3 \times \mathbb{Z}_2^+, & \mathbb{Z}_2^+ &= \{0, 1\}, & \text{addition} \\ \implies & \text{exercise: Show } D_6 = D_3 \times \mathbb{Z}_2^+. \end{aligned}$$